MA416

Lecture 06 – September 18, 2025
Pairwise Comparison Tests

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Tests on One Variance: $H_0: \sigma^2 = \sigma_0^2$

Theorem 1: Suppose that $X_k \sim^{iid} N(\mu, \sigma^2)$ with $H_0: \sigma^2 = 2$ and let S_x^2 be the associated sample variance for a sample of size n. Then if the null hypothesis is true, then:

$$\chi^2_{stat} := \frac{(n-1) \cdot S_x^2}{\sigma_0^2} \sim \chi^2_{n-1}$$

Example (test statistic approach):

```
S = c(5, 17, 7, 14, 19, 18, 21, 5, 6)
sx = sd(S)
sx2 = var(S)

sig2_0 = 49

c(sig2_0, sx2)
sqrt(c(sig2_0, sx2))
```

```
> c(sig2_0, sx2)
[1] 49.00000 44.02778
> sqrt(c(sig2_0, sx2))
[1] 7.000000 6.635343
```

```
n = length(S)
chisq2_stat  = (n-1)*var(S)/sig2_0

alpha = 0.05
c2crit_L = qchisq(alpha/2, n-1, lower.tail=T)
c2crit_R = qchisq(alpha/2, n-1, lower.tail=F)

c(c2crit_L, c2crit_R)
chisq2_stat
```

```
> c(c2crit_L, c2crit_R)
[1]  2.179731 17.534546
> chisq2_stat
[1] 7.188209
```

Be able to interpret and statistically conclude!

Tests on One Variance: $H_0: \sigma^2 = \sigma_0^2$

Theorem 2: Suppose that $X_k \sim^{iid} N(\mu, \sigma^2)$ with $H_0: \sigma^2 = 2$ and let S_x^2 be the associated sample variance for a sample of size n. Then:

$$(CI)_{\sigma^2}^{c\%} = \left(\frac{(n-1) \cdot s_x^2}{\chi^2_{crit_R}}, \frac{(n-1) \cdot s_x^2}{\chi^2_{crit_L}} \right)$$

```
> c(sig2_0, sx2)
[1] 49.00000 44.02778
> sqrt(c(sig2_0, sx2))
[1] 7.000000 6.635343
```

```
> c(c2crit_L, c2crit_R)
[1]  2.179731 17.534546
> chisq2_stat
[1] 7.188209
```

Example (p-value approach):

```
chisq2_stat_L  = (n-1)*var(S)/sig2_0
chisq2_stat_R = (n-1)*sig2_0/var(S)

# definitions on L/R depending on size comparisons :)
c(chisq2_stat_L, chisq2_stat_R)

pval = pchisq(chisq2_stat_L, n-1, lower.tail=T) +
              pchisq(chisq2_stat_R, n-1, lower.tail=F)
pval
```

```
> pval
[1] 0.8340351
```

Example (C.I. approach):

```
ci_var = (n-1)*var(S)*c(1/c2crit_R, 1/c2crit_L)
ci_sd = sqrt(ci_var)
```

```
> ci_var
[1]  20.08733 161.58978
> ci_sd
[1]  4.481889 12.711797
```

Be able to interpret and statistically conclude!

Tests on Two Variances: $H_0: \sigma_1^2 = \sigma_2^2$

Theorem 3: Suppose that $X_{k_1} \sim^{iid} N(\mu_1, \sigma_1^2)$ and $X_{k_2} \sim^{iid} N(\mu_2, \sigma_2^2)$ with $\text{Cov}(X_{k_1}, X_{k_2}) = 0$ and $H_0: \sigma_1^2 = \sigma_2^2$ being true. Then if S_1^2, S_2^2 are the associated sample variances, then:

$$F_{stat} := \frac{S_1^2}{S_2^2} \sim F_{(n_1-1, n_2-1)}$$

Example:

```
set.seed(123)
S1 = runif(32, 4, 19)
S2 = rexp(23, 1/7.7)

var1 = var(S1)
var2 = var(S2)

c(var1, var2)
sqrt(c(var1, var2))

fstat = var2/var1
fstat
```

```
> c(var1, var2)
[1] 19.64806 44.84728
> sqrt(c(var1, var2))
[1] 4.432613 6.696811
> fstat
[1] 2.28253
```

```
alpha = 0.05
fcrit = qf(alpha, length(S2)-1, length(S1)-1, lower.tail=F)
pval = pf(fstat, length(S2)-1, length(S1)-1, lower.tail=F)
```

```
> fcrit
[1] 1.895552
> pval
[1] 0.01719413
```

Be able to interpret and statistically conclude!

Tests on $g \geq 3$ Variances: $H_0: \sigma_1^2 = \sigma_2^2 = \cdots = \sigma_g^2$

Without creating "new" statistical tools, we can test the equality of them in a pairwise fashion by using "pairwise comparison" tests (since we already know how to test two-variance equality).

Note: For $g = 4$, If $\sigma_1^2 = \sigma_2^2 = \sigma_3^2 = \sigma_4^2$, then all the following are true:

$$\sigma_1^2 = \sigma_2^2 \qquad \sigma_1^2 = \sigma_3^2 \qquad \sigma_1^2 = \sigma_4^2$$
$$\sigma_2^2 = \sigma_3^2 \qquad \sigma_2^2 = \sigma_4^2$$
$$\sigma_3^2 = \sigma_4^2$$

Which is a total of $3 + 2 + 1 = C_2^4 = \frac{4!}{2! \cdot 2!} = 6$ pairwise comparison tests to conduct.

Important: If we reject at least one $H_0^{i,j}: \sigma_i^2 = \sigma_j^2$, then we will reject $H_0: \sigma_1^2 = \sigma_2^2 = \cdots = \sigma_g^2$.

Tests on $g \geq 3$ Variances: $H_0: \sigma_1^2 = \sigma_2^2 = \cdots = \sigma_g^2$

Example:

```
set.seed(123)
S1 = rnorm(14, 121, 3)
S2 = rnorm(15, 124, 3)
S3 = rnorm(17, 127, 3)

n1 = length(S1)
n2 = length(S2)
n3 = length(S3)

sdx = c(sd(S1), sd(S2), sd(S3))
sdx
```

```
> sdx
[1] 2.560089 3.036005 2.749588
```

```
fstat_12 = var(S2)/var(S1)
fstat_13 = var(S3)/var(S1)
fstat_23 = var(S2)/var(S3)

c(fstat_12, fstat_13, fstat_23)

pval_12 = pf(fstat_12, n2-1, n1-1, lower.tail=F)
pval_13 = pf(fstat_13, n3-1, n1-1, lower.tail=F)
pval_23 = pf(fstat_23, n2-1, n3-1, lower.tail=F)

c(pval_12, pval_13, pval_23)
```

```
> c(fstat_12, fstat_13, fstat_23)
[1] 1.406354 1.153520 1.219185
> c(pval_12, pval_13, pval_23)
[1] 0.2725775 0.4025777 0.3487430
```

Be able to interpret and statistically conclude with $\alpha_{all} = 0.25, 0.30, 0.45$, and $\alpha_{12} \neq \alpha_{13} \neq \alpha_{23}$.